

HOLLISTER INCORPORATED

PLC Harmonization & Semantic Integration Strategy (DRAFT)

Unified Architecture for Critical Manufacturing MES,
Mendix Composability Layer, Altair Federated Ontology Fabric,
and SAP S/4HANA Enterprise Integration

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Ontology-Driven Integration, Low-Code Composability, and Enterprise Data Fabric

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1. Executive Summary

This document defines the updated PLC Harmonization Strategy for Hollister Incorporated, evolving the baseline architecture established in HOLL-ARCH-001 and the technical integration specification in HOLL-TECH-PLC-MES-001. The strategy introduces three transformative architectural capabilities that elevate the Digital Manufacturing Footprint from a well-instrumented factory floor to a semantically governed, composable, and AI-ready manufacturing intelligence platform.

Critical Manufacturing MES serves as the validated Engine of Record for cGMP-compliant production execution across all Hollister global facilities. **Mendix** provides a Low-Code Agility Layer that enables rapid development of persona-specific applications without disturbing the validated MES core. **Altair RapidMiner with Graph Studio** delivers a Federated Ontology Fabric that unifies the semantic knowledge layer across equipment hierarchies, process parameters, quality attributes, and supply chain entities, providing the connective tissue that makes the entire architecture self-describing and machine-interpretable.

The integration with SAP S/4HANA completes the enterprise circuit, ensuring that the semantic knowledge layer bridges the traditional OT-IT divide and extends into financial planning, materials management, and supply chain optimization. Together, these components create a unified architecture where every PLC tag, every sensor reading, every batch record, and every quality event exists within a semantically rich context that enables both human understanding and AI-driven decision support.

Strategic Outcome: A manufacturing intelligence platform where the ontology fabric acts as the universal translation layer between shop floor automation, production execution, enterprise resource planning, and advanced analytics, eliminating the brittle point-to-point integrations that constrain agility in regulated manufacturing.

2. Architectural Vision & Governing Principles

2.1 The Four-Pillar Architecture

The harmonized architecture rests on four interdependent pillars, each serving a distinct role while sharing a common semantic foundation:

Pillar	Platform	Role	Governance Scope
Engine of Record	Critical Manufacturing MES	Validated production execution, eDHR, batch management, quality enforcement	FDA 21 CFR Part 11, ISO 13485, QMSR, EU MDR
Agility Layer	Mendix (Siemens Xcelerator)	Low-code composable applications, persona dashboards, semantic search, workflow orchestration	Innovation velocity without core re-validation
Ontology Fabric	Altair RapidMiner + Graph Studio	Federated knowledge graph, semantic governance, canonical data model, Graph-RAG enablement	Data lineage, vocabulary governance, and cross-domain semantic consistency
Enterprise Backbone	SAP S/4HANA	Financial accounting, materials management, supply chain planning, master data	Enterprise master data authority, financial compliance

2.2 Governing Principles (Updated)

The following principles extend the original HOLL-ARCH-001 design principles with semantic governance and composability dimensions:

Principle	Description	Compliance Mapping
Semantic First	Every data element carries machine-interpretable meaning through ontological classification. Raw PLC tags are never exposed to consuming applications without semantic context.	ISA-95/IEC 62264, OPC-UA Information Models
Composability	Business capabilities are assembled from modular, reusable components rather than monolithic applications. Mendix apps consume canonical services without coupling to physical infrastructure.	Siemens Xcelerator, Microservices Architecture
Ontology as Contract	The federated ontology in Altair Graph Studio serves as the binding contract between data producers and consumers, replacing brittle schema mappings with semantic relationships.	W3C OWL/RDFS, SKOS, ISA-95 B2MML

Validated Core, Agile Edge	The Critical Manufacturing MES core remains change-controlled and validated. Innovation occurs in the Mendix satellite layer, which reads from but does not write to validated systems without explicit governance.	GAMP 5, FDA 21 CFR Part 11, ISO 13485
Federated Governance	No single system owns all metadata. The ontology fabric federates knowledge across MES, ERP, PLM, and IoT, with Altair Graph Studio as the reconciliation engine.	DAMA DMBOK, Enterprise Knowledge Graph patterns
Human-in-the-Loop AI	All AI/ML recommendations from the semantic layer require human acknowledgment via digitally signed workflows before affecting validated production systems.	FDA AI/ML guidance, ICH Q10, QMSR

3. PLC Harmonization Architecture

3.1 Vendor-Neutral Abstraction (Evolved)

The PLC harmonization strategy defined in HOLL-TECH-PLC-MES-001 established a six-layer integration architecture from physical devices to application consumers. This updated strategy preserves the proven layered model while enhancing three critical capabilities: the semantic enrichment of all PLC data through the Altair ontology fabric, the decoupling of application development from integration complexity through Mendix composability, and the bidirectional synchronization with SAP S/4HANA to maintain enterprise master data authority.

3.2 Tag-to-Ontology Mapping Pipeline

The original tag naming convention (Site_Area_Equipment_Module_Point_Attribute) is preserved as the physical layer identifier. The ontology fabric introduces a semantic layer that maps every physical tag to one or more ontological concepts, enabling cross-domain reasoning that was impossible with flat tag hierarchies alone.

Layer	Identifier	Example	Maintained In
Physical Tag	PLC tag per naming convention	BAL_MH1_MOLD01_HYD_PRESSURE_PV	PLC Program / AssetCentre
Canonical Name	Vendor-neutral functional name	CycleTime, MeltTemperature	Metadata Registry (Edge Layer)
Ontology Concept	Semantic class + relationships	holl: HydraulicPressure rdfs:subClassOf holl: ProcessParameter	Altair Graph Studio
Enterprise Entity	SAP master data reference	Material HOLL-CATH-2400, Equipment BAL-MOLD-001	SAP S/4HANA MDG

3.3 PackML State Model Enhancement

The PackML/ISA-88 state model defined in the PLC-MES specification is extended with ontological state annotations. Each PackML state transition is mapped to ontology events that trigger downstream reasoning in the knowledge graph. For example, a transition from Execute (State 6) to Holding (State 10) not only updates the Critical Manufacturing MES resource state but also triggers the ontology fabric to evaluate related quality holds, upstream material genealogy, and operator qualification status in a single graph traversal.

3.4 PLC Platform Consolidation Roadmap

Consistent with the baseline architecture, the Rockwell Automation ControlLogix/CompactLogix/GuardLogix family remains the primary PLC platform. The

harmonization strategy adds explicit interoperability requirements for Siemens, Schneider, Beckhoff, and Omron platforms through the edge abstraction layer. The ontology fabric provides the semantic glue that makes multi-vendor environments appear uniform to all consuming applications.

Vendor	Platform	Edge Protocol	Ontology Connector	SAP Integration
Rockwell	ControlLogix 5580	EtherNet/IP via Kepware	Native OPC-UA info model	PP/DS via CM MES
Siemens	S7-1500	PROFINET via Kepware	OPC-UA Machinery Companion	Direct S7-S/4 connector
Schneider	Modicon M580	Modbus TCP / EtherNet/IP	Custom adapter + registry	Via integration platform
Beckhoff	TwinCAT 3	Native OPC-UA	ADS + OPC-UA model	Via integration platform
Omron	NX/NJ Series	EtherNet/IP / EtherCAT	Custom adapter + registry	Via integration platform

4. Critical Manufacturing MES: Engine of Record

4.1 Platform Positioning

Critical Manufacturing (CM) serves as Hollister's validated Engine of Record for all production execution activities. As a cloud-native, HTML5-based MES platform, CM provides the regulated backbone for electronic Device History Records (eDHR), batch management, material genealogy, and quality enforcement. The platform's Connect IoT module provides the native bridge between the PLC edge layer and the MES application tier, consuming the canonical data model established in the integration specification.

4.2 CM Core Functions (Protected Scope)

The following CM core functions are maintained under strict change control and validation governance. These represent the validated state that Mendix satellite applications must not modify directly:

CM Module	Function	Validation Level
Electronic DHR	Complete electronic Device History Record with digital signatures, material traceability, and process parameter capture	Full GAMP 5 Cat 5; 21 CFR Part 11 validated
Connect IoT	Real-time equipment integration, OPC-UA/MQTT ingestion, equipment state management, process data collection	GAMP 5 Cat 4; IQ/OQ/PQ per equipment
fabLIVE	Real-time factory dashboard, OEE calculation, production monitoring, alarm visualization	GAMP 5 Cat 4; validated calculations
Material Management	Lot genealogy, material consumption tracking, material quarantine/release, expiry management	GAMP 5 Cat 5; full traceability validation
Quality Management	In-process inspection, SPC, non-conformance handling, CAPA integration, parametric data collection	GAMP 5 Cat 5; 21 CFR Part 11 validated
Recipe Management	ISA-88 recipe lifecycle, recipe versioning, equipment-recipe qualification, parameter download to PLCs	GAMP 5 Cat 5; full validation required

4.3 CM-to-Ontology Bridge

Critical Manufacturing exposes its data model through RESTful APIs and event-driven webhooks. The ontology fabric consumes CM events (state changes, production completions, quality inspections, material movements) and enriches them with semantic context from the

Altair knowledge graph. This enables the Mendix agility layer to present semantically rich views of production data without requiring direct CM API knowledge.

Data Flow Pattern: CM Production Event → Kafka Topic (production.events.{site}) → Ontology Enrichment Service (Altair) → Enriched Event on Kafka → Mendix Application Consumer. This pattern ensures that the ontology acts as a passive enrichment layer that does not impede the real-time validated data flow from CM.

5. Mendix Composability Layer: Agility Without Risk

5.1 Strategic Rationale

In a cGMP-regulated medical device environment, modifying core MES workflows triggers extensive re-validation cycles that can span weeks to months. Mendix, as part of the Siemens Xcelerator ecosystem, provides a low-code platform that enables rapid development of satellite applications that consume MES data and ontology services without altering the validated CM core. This creates an innovation velocity multiplier: the core remains stable and validated while the edge applications evolve rapidly to meet emerging operational needs.

5.2 Mendix Architectural Roles

5.2.1 Ontology Visualization & Semantic Search

Mendix serves as the primary front-end for the Altair knowledge graph, providing persona-specific views into the semantic data layer. While the Altair Graph Studio stores and maintains the ontology relationships, Mendix renders these relationships as intuitive dashboards tailored to specific user roles:

Persona	Mendix Dashboard	Ontology Query Pattern	Data Sources
Quality Engineer	Risk View: Visual risk graph showing CPP-CQA relationships, control limits, and trending deviations across related equipment	Graph traversal: CQA → controlledBy → CPP → measuredBy → Sensor → calibratedPer → Standard	CM Quality, Altair Graph, Calibration DB
Process Engineer	Genealogy View: Complete material and process genealogy with drill-down to individual sensor traces and parametric data	Graph traversal: Batch → usedMaterial → Lot → suppliedBy → Supplier, intersected with Equipment → ranRecipe → Parameters	CM eDHR, CM Material, Altair Graph
Maintenance Tech	Semantic Maintenance Advisor: Natural language search across equipment manuals, maintenance histories, and failure modes using Graph-RAG	Graph-RAG: Query → Entity extraction → Graph context retrieval → LLM-augmented response with traceable citations	Altair Graph, CM Connect IoT, Document Store
Plant Manager	Single Pane of Glass: Unified view of OEE (from fabLIVE), cost-per-unit (from SAP), quality yield (from CM), and predictive maintenance alerts	Federated query: Site → hasLine → Line → currentOEE + costPerUnit + yieldRate + maintenanceRiskScore	CM fabLIVE, SAP S/4HANA, Altair Graph

Supply Chain	Demand-Supply Intelligence: Real-time production capacity linked to demand signals, inventory positions, and supplier risk scores	Graph traversal: Product → requiredMaterial → inventoryAt → Site, intersected with SupplierRisk → alternateSource	SAP S/4HANA, CM Production, Altair Graph
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5.2.2 The Validation Gap Bridge

Mendix addresses the validation gap through a strict architectural boundary: Mendix applications operate in read-only mode against the CM-validated core, consuming data through the Kafka event stream and the ontology enrichment layer. When a Mendix application needs to trigger a validated action (such as an operator's digital signature or a quality hold), it does so through CM's official API gateway, which enforces all validation and Part 11 compliance controls.

This pattern enables three categories of satellite applications:

Category	Description	Validation Requirement
Read-Only Dashboards	Visualize production, quality, and maintenance data from CM and SAP with ontology-enriched context. No write-back to validated systems.	Minimal: data accuracy verification against source systems. No Part 11 validation required.
Workflow Orchestrators	Coordinate human-in-the-loop decision processes (shift handover, changeover checklists, deviation triage) that ultimately trigger CM API calls for validated actions.	Moderate: workflow logic verification; CM API call validation handled by CM itself.
AI-Assisted Advisors	Graph-RAG powered applications that provide recommendations (root cause, predictive maintenance, recipe optimization) requiring human approval before execution.	Moderate-to-High: AI model validation, recommendation accuracy testing, human-in-the-loop verification.

5.2.3 Siemens Xcelerator Synergies

Mendix's position within the Siemens Xcelerator ecosystem creates strategic interoperability advantages. Critical Manufacturing is a Siemens portfolio company, ensuring tight integration between CM's Connect IoT module and Mendix's integration capabilities. This alignment simplifies the documentation burden for ISO 13485 and QMSR compliance, as the platform interoperability has been validated within the Siemens ecosystem rather than requiring custom integration validation between independent vendors.

6. Federated Ontology Fabric: Altair RapidMiner & Graph Studio

6.1 Why Altair for the Ontology Fabric

The selection of Altair RapidMiner with Graph Studio as the ontology platform reflects a strategic convergence of three capabilities: enterprise-grade knowledge graph management (Graph Studio), integrated AI/ML pipelines for predictive analytics and anomaly detection (RapidMiner), and federated data integration across heterogeneous sources. This combination enables Hollister to build, maintain, and operationalize a federated ontology that spans the full manufacturing value chain from PLC sensor data to SAP financial transactions.

6.2 Ontology Architecture

6.2.1 Core Ontology Domains

The federated ontology is organized into domain-specific sub-ontologies that are composed into a unified knowledge graph through shared upper ontology patterns:

Domain Ontology	Scope	Key Classes	Source Systems
holl:Equipment	Equipment hierarchy, capabilities, states, maintenance schedules, calibration status	EquipmentClass, WorkCenter, ControlModule, Sensor, Actuator	CM Connect IoT, PLC Registry, CMDB
holl:Process	Process parameters, recipes, control strategies, SPC rules, CPP-CQA relationships	ProcessParameter, Recipe, ControlStrategy, CriticalAttribute	CM Recipe Mgmt, CM Quality, DCS
holl:Material	Material definitions, lot genealogy, specifications, supplier qualification	Material, MaterialLot, Specification, SupplierQualification	CM Material, SAP MM, PLM
holl:Quality	Quality events, inspections, deviations, CAPA, risk assessments	InspectionResult, Deviation, CAPA, RiskAssessment, ControlChart	CM Quality, QMS, Regulatory
holl:Product	Product definitions, BOM, design history, regulatory submissions	ProductDefinition, BillOfMaterials, DesignHistory, RegulatoryDossier	PLM (Aras Interim), SAP PP
holl:Supply	Supply chain entities, demand signals, inventory positions, logistics	SupplyChainNode, DemandSignal, InventoryPosition, TransportLeg	SAP S/4HANA SCM, CM Production

6.2.2 Upper Ontology Alignment

All domain ontologies inherit from a shared upper ontology aligned with established standards. ISA-95 (IEC 62264) provides the equipment hierarchy and operations management model. ISA-88 provides the procedural control model for recipes and batch management. OPC-UA companion specifications provide the information model alignment for PLC data. The W3C Semantic Sensor Network (SSN) ontology provides the sensor observation model. PROV-O provides the provenance model for data lineage and audit trails. This standards alignment ensures that the Hollister ontology is interoperable with industry tools and can be extended as new standards emerge.

6.2.3 Graph-RAG for Manufacturing Intelligence

The Altair platform's integration of RapidMiner AI/ML pipelines with Graph Studio's knowledge graph enables a Graph-RAG (Retrieval-Augmented Generation) architecture specifically designed for manufacturing contexts. When an operator or engineer queries the Mendix-based Semantic Maintenance Advisor, the query flows through a pipeline that extracts entities from the natural language question, maps those entities to ontology concepts, traverses the knowledge graph to retrieve relevant context (equipment specifications, maintenance history, similar failure modes across plants, related quality deviations), and feeds this structured context to a language model for response generation. The response includes traceable citations back to the knowledge graph, ensuring that every recommendation can be audited.

6.3 Federated Data Integration Pattern

The ontology fabric operates as a federated system, meaning it does not replicate all data into a central store. Instead, it maintains a knowledge graph of relationships and metadata that points to authoritative data in source systems. When a Mendix application or an AI/ML pipeline needs to resolve a query, the ontology provides the semantic map while the underlying data is fetched from CM, SAP, or the historian at query time.

Federation Pattern: Altair Graph Studio stores ontology instances (equipment definitions, relationships, metadata) as persistent graph data. Time-series process data remains in the FactoryTalk Historian or OSIsoft PI. Transactional data remains in CM and SAP. The ontology provides the query routing intelligence that knows which system holds the authoritative answer for any given question.

7. SAP S/4HANA Enterprise Integration

7.1 Integration Architecture

SAP S/4HANA serves as the enterprise backbone for financial accounting, materials management, production planning, and supply chain management. The integration with the PLC-MES-Ontology architecture follows a bidirectional pattern mediated by the Kafka event streaming platform and the ontology enrichment layer.

Integration Point	Direction	Data Flow	SAP Module	Mediator
Work Order	SAP → CM	Production order release with BOM, routing, and quantity triggers recipe download to PLC layer	PP/DS	Kafka + Ontology
Goods Receipt	CM → SAP	Batch completion in CM triggers goods receipt posting in SAP with quality status and quantity	MM/MM	Kafka + API
Material Master	SAP → Graph	Material master data changes propagate to ontology fabric for cross-system consistency	MDG	SAP CPI + Graph API
Cost Allocation	CM → SAP	Production actuals (machine time, material consumption, scrap) feed CO module for cost-per-unit	CO/PA	Kafka + API
Quality Notify	CM → SAP	Quality deviations in CM create QM notifications in SAP for enterprise CAPA tracking	QM	Kafka + API
Supply Chain	SAP ↔ Graph	Demand signals from SAP APO intersected with production capacity from CM through ontology-mediated planning	APO/IBP	Ontology Federation

7.2 Master Data Governance with Ontology

SAP S/4HANA Master Data Governance (MDG) remains the authoritative source for enterprise master data (materials, equipment, business partners). The ontology fabric in Altair Graph Studio consumes SAP master data through the SAP Cloud Platform Integration (CPI) connector and enriches it with manufacturing-specific semantic relationships that SAP does not natively model. For example, SAP knows that Material HOLL-CATH-2400 exists and has a BOM. The ontology knows that this material requires specific CPPs on injection molding equipment, is sensitive to humidity variations beyond a defined threshold, and has historically shown higher

defect rates when produced on Line 3 during third shift. This semantic enrichment transforms static master data into actionable manufacturing intelligence.

7.3 Financial-Operational Convergence

The integration architecture enables a financial-operational convergence view in Mendix that was previously impossible with siloed systems. The Plant Manager dashboard in Mendix pulls real-time OEE from CM fabLIVE, links it to cost-per-unit data from SAP CO/PA, overlays quality yield from CM Quality, and contextualizes all of this with predictive maintenance risk scores from the Altair RapidMiner ML models. This unified view is assembled through ontology-mediated queries that traverse the knowledge graph to correlate data from systems that have no direct integration with each other.

8. End-to-End Integration Topology

8.1 Layered Data Flow

The complete integration topology extends the original six-layer model from HOLL-TECH-PLC-MES-001 with explicit ontology and composability layers:

Layer	Name	Components	Enhancement in v2.0
L1	Physical Device	PLCs (Rockwell, Siemens, Schneider, Beckhoff, Omron), sensors, actuators, vision systems	Ontology-aligned tag naming; PackML state annotations
L2	Communication	OPC-UA servers, MQTT brokers, EtherNet/IP, PROFINET	OPC-UA Companion Spec alignment with ontology info models
L3	Edge Computing	Kepware, Ignition Edge, Azure IoT Edge, EdgeX Foundry	Edge-level ontology enrichment; local Graph-RAG cache
L4	Semantic/Context	Altair Graph Studio (knowledge graph), RapidMiner (AI/ML), Metadata Registry	NEW: Federated ontology fabric replacing flat canonical model
L5	Integration Platform	Apache Kafka, Schema Registry, API Gateway	Ontology-enriched event streams; semantic routing
L6	Application	Critical Manufacturing MES, SAP S/4HANA, Mendix apps, Digital Twin, Analytics	NEW: Mendix composability layer consuming ontology services

8.2 Kafka Topic Structure (Extended)

The Kafka topic structure from HOLL-TECH-PLC-MES-001 is extended with ontology enrichment topics that carry semantically annotated events:

Topic Pattern	Purpose	Consumer
equipment.state.{site}	Raw PackML state changes from edge	CM Connect IoT, Ontology Enrichment
production.events.{site}	Cycle complete, material consumed	CM MES Adapter, Ontology Enrichment
ontology.enriched.equipment.{site}	Semantically annotated equipment events with graph context	Mendix Apps, Digital Twin, Analytics
ontology.enriched.quality.{site}	Quality events with CPP-CQA relationships and risk context	Mendix Quality Dashboard, CM QM

ontology.enriched.maintenance.{site}	Maintenance events with failure mode genealogy and predictive scores	Mendix Maintenance Advisor, CMMS
sap.masterdata.changes	SAP master data change events for ontology synchronization	Altair Graph Studio, CM Master Data
sap.production.orders	Production order releases from SAP PP/DS	CM Recipe Manager, Ontology Fabric

9. Security, Compliance & Governance Architecture

9.1 Regulatory Compliance Mapping (Extended)

The addition of the ontology fabric and Mendix composability layer introduces new compliance considerations that extend the baseline HOLL-ARCH-001 regulatory mapping:

Regulation	New Requirements from v2.0 Architecture	Implementation
FDA 21 CFR Part 11	Ontology-enriched data must maintain audit trail continuity. Mendix apps that trigger validated actions must pass through CM's Part 11 controls.	Ontology enrichment is additive (append-only); all CM API calls from Mendix authenticated with Part 11 e-signatures.
FDA AI/ML Guidance	Graph-RAG recommendations must be traceable and explainable. Human-in-the-loop required for all AI-influenced production decisions.	Altair RapidMiner model registry with version control; Mendix workflow for human approval before AI recommendations reach CM.
QMSR (replacing QSR)	Quality management system must demonstrate process understanding enabled by semantic layer.	Ontology-driven CPP-CQA relationship management; automated impact assessment for process changes via graph traversal.
EU MDR 2017/745	UDI and post-market surveillance data must be accessible through the unified architecture.	SAP-CM-Ontology integration for complete UDI lifecycle; Mendix PMS dashboard consuming enriched quality data.
GDPR	Ontology must not store PII beyond processing purpose. Mendix apps consuming operator data must comply with data minimization.	Operator identity stored only in CM (validated); ontology references anonymized operator roles, not individuals.
IEC 62443	Ontology fabric and Mendix layer must operate within defined security zones. Graph queries must not traverse zone boundaries without firewall mediation.	Altair Graph Studio deployed in Level 4 (IT network); Mendix in Level 4/5; all OT data flows through L3.5 DMZ.

9.2 Data Integrity (ALCOA+ in the Semantic Layer)

The ALCOA+ principles established in HOLL-ARCH-001 are extended to cover the ontology fabric. All ontology enrichment operations are treated as non-destructive, append-only transformations that preserve the original data integrity of source records. Ontology annotations carry their own provenance metadata (who enriched, when, which ontology version, which rules applied), ensuring that the semantic layer enhances rather than obscures the ALCOA+ chain. The Altair Graph Studio maintains a complete version history of the ontology schema itself, ensuring that historical data can always be interpreted in the context of the ontology version that was active at the time of production.

10. Implementation Roadmap

10.1 Phased Deployment Plan

The implementation roadmap aligns with the HOLL-ARCH-001 phased deployment while introducing the new architectural components at strategic insertion points:

Phase	Timeline	Scope	Key Deliverables
Phase 0: Foundation	Months 1-3	Architecture validation, vendor selection (Altair, Mendix licensing), ontology domain modeling, CoE establishment	Approved architecture; ontology domain models for Equipment and Process; Mendix environment provisioned; CoE charter
Phase 1: Pilot Line + Ontology	Months 4-9	Single line at Ballina fully instrumented with CM MES; Altair Graph Studio deployed with Equipment and Process ontologies; first Mendix dashboard (fabLIVE + ontology)	Baseline line operational; ontology fabric live with 2 domain ontologies; Mendix Plant Manager Single Pane of Glass prototype
Phase 2: Semantic Maintenance Pilot	Months 10-14	Graph-RAG Semantic Maintenance Advisor built in Mendix for pilot line; RapidMiner predictive maintenance models trained; SAP work order integration enabled	Semantic Maintenance Advisor in production on 1 line; measurable KPI: 30% faster root cause identification
Phase 3: Ballina Full + SAP Deep	Months 15-24	All Ballina lines on CM MES with ontology enrichment; full SAP S/4HANA bidirectional integration; Material and Quality ontology domains activated	All 6-8 lines operational; full SAP integration; 6 domain ontologies active; Mendix app catalog with 5+ satellite apps
Phase 4: Second Plant	Months 18-30	Replicate to second plant (Libertyville recommended for Maintenance Advisor impact measurement); cross-plant ontology federation enabled	Second plant operational; cross-plant analytics via federated ontology; Mendix apps deployed to second plant
Phase 5: Enterprise	Months 24-48	All plants connected through federated ontology fabric; enterprise supply chain optimization enabled; permanent PLM integration replacing Aras interim	Enterprise-wide semantic manufacturing intelligence platform; Supply Chain Control Tower in Mendix

10.2 Low-Code Innovation Lab (Days 61-90 CoE Initiative)

As part of the Center of Excellence establishment in Phase 0, a Low-Code Innovation Lab is recommended to demonstrate the value of the Mendix + Ontology architecture rapidly and tangibly:

1. **Pilot Application:** Build a Semantic Maintenance Advisor for one production line at the Libertyville facility using Mendix, connected to the Critical Manufacturing Connect IoT layer and the Altair Knowledge Graph.
2. **Integration Logic:** The Mendix app consumes equipment state events from Kafka, enriches them through the Altair Graph Studio API, and presents maintenance technicians with contextual recommendations that traverse the equipment-failure mode-maintenance history-spare parts-supplier ontology path.
3. **Target Outcome:** Demonstrate that operators can identify root causes 30% faster using semantic relationships than by searching the standalone MES or consulting PDF manuals. This measurable KPI justifies the broader ontology fabric investment.
4. **Success Criteria:** Mean time to root cause reduction; operator satisfaction survey; reduction in repeat maintenance events; demonstrated Graph-RAG accuracy against known failure scenarios.

11. Investment Estimate (Order of Magnitude)

The following ROM estimate extends the HOLL-ARCH-001 investment model to include the new architectural components. Existing estimates for PLCs, sensors, DCS, cameras, network, and compute infrastructure remain as per HOLL-ARCH-001 and are not repeated here.

Component	Pilot (1 Line)	Full Plant	Notes
Critical Manufacturing MES	\$400K - \$600K	\$1.5M - \$2.5M	CM licensing, Connect IoT, fabLIVE, eDHR
Mendix Platform	\$120K - \$200K	\$350K - \$600K	Mendix licenses (Siemens Enterprise pricing available)
Altair RapidMiner + Graph Studio	\$150K - \$250K	\$400K - \$700K	Platform licensing, ontology development, Graph-RAG setup
SAP Integration (CPI + Custom)	\$100K - \$180K	\$300K - \$500K	SAP CPI connectors, custom adapters, MDG configuration
Apache Kafka Cluster	\$50K - \$100K	\$150K - \$250K	Confluent Cloud or on-prem cluster with Schema Registry
Ontology Engineering	\$200K - \$350K	\$400K - \$600K	Domain modeling, taxonomy development, mapping services
Mendix App Development	\$150K - \$250K	\$500K - \$900K	5-8 satellite applications across persona dashboards
Systems Integration	\$300K - \$500K	\$1.2M - \$2.0M	End-to-end integration, testing, deployment
Validation (IQ/OQ/PQ)	\$200K - \$350K	\$800K - \$1.2M	CM validation, ontology V&V, Mendix app qualification
TOTAL (New Components)	\$1.67M - \$2.78M	\$5.6M - \$9.25M	Additive to HOLL-ARCH-001 infrastructure estimate

Combined Investment (Infrastructure + Semantic Platform): When combined with the HOLL-ARCH-001 estimates (\$2.2M-\$3.5M pilot line; \$14.9M-\$22.6M full plant), the total investment for a fully instrumented, semantically governed, composable manufacturing intelligence platform ranges from \$3.9M-\$6.3M for the pilot line to \$20.5M-\$31.9M for a full 8-line plant deployment. The incremental investment in the semantic and composability layers represents approximately 40% of the total, delivering capabilities that fundamentally transform the value extracted from the baseline automation investment.

12. Strategic Recommendations

12.1 Immediate Actions (Days 1-30)

1. **Secure Altair RapidMiner + Graph Studio evaluation licenses** and begin ontology domain modeling for the Equipment and Process domains using the existing PLC tag registry from HOLL-TECH-PLC-MES-001 as the seed data.
2. **Engage Critical Manufacturing professional services** for Connect IoT architecture review aligned with the updated semantic integration requirements.
3. **Provision Mendix development environment** through Siemens Xcelerator licensing and identify 2-3 citizen developers for the Low-Code Innovation Lab.
4. **Initiate SAP CPI connector assessment** for master data synchronization between S/4HANA MDG and the Altair Graph Studio.

12.2 Medium-Term (Days 31-90)

1. **Establish the Low-Code Innovation Lab within the CoE** with the Semantic Maintenance Advisor as the flagship pilot application.
2. **Complete ontology domain models** for Equipment, Process, and Material domains with formal review by Engineering, Quality, and Manufacturing stakeholders.
3. **Deploy the Kafka event streaming cluster** with the extended topic structure supporting ontology enrichment streams.
4. **Validate the CM-to-Ontology-to-Mendix data flow pattern** on a single equipment instance to prove the end-to-end architecture before scaling.

12.3 Long-Term Vision

The end-state vision is a self-describing manufacturing enterprise where every asset, process, material, and quality event exists within a semantically rich knowledge fabric that enables both human understanding and autonomous AI-driven optimization. The federated ontology in Altair Graph Studio becomes the institutional memory of the manufacturing organization, capturing not just what happened but why it happened, what it means, and what should happen next. Mendix ensures this intelligence is accessible to every persona from the shop floor to the boardroom, while Critical Manufacturing and SAP S/4HANA ensure that every action taken on this intelligence is validated, compliant, and traceable.

This document is a living strategy specification. It will be updated as the architecture progresses through Phase 0, detailed engineering, and vendor selection.